

Shape-Constrained Additive Models

Simon Frost

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Introduction

Standard GAMs estimate smooth functions without restrictions on their shape. In many applications, however, domain knowledge dictates that a relationship should be monotonically increasing (e.g., dose-response), convex (e.g., cost curves), or satisfy other shape constraints.

Shape-Constrained Additive Models (SCAMs) enforce these constraints using SCOP-splines (Shape Constrained P-splines), where B-spline coefficients are reparameterized through cumulative sums and exponentiation to guarantee the desired shape.

GAM.jl implements SCAM fitting following the approach of the R [scam](#) package (Pya & Wood, 2015).

Setup

```
using GAM
using StatsAPI: predict
using DataFrames
using CSV
using Random
using Statistics
using Plots
```

Available basis types

GAM.jl supports 8 shape-constrained basis types, specified via the `bs` argument in `s()`:

bs symbol	Constraint	Description
:mpi	Monotone increasing	$f'(x) \geq 0$
:mpd	Monotone decreasing	$f'(x) \leq 0$
:cx	Convex	$f''(x) \geq 0$
:cv	Concave	$f''(x) \leq 0$
:micx	Monotone increasing & convex	$f'(x) \geq 0$ and $f''(x) \geq 0$
:micv	Monotone increasing & concave	$f'(x) \geq 0$ and $f''(x) \leq 0$
:mdcx	Monotone decreasing & convex	$f'(x) \leq 0$ and $f''(x) \geq 0$
:mdcv	Monotone decreasing & concave	$f'(x) \leq 0$ and $f''(x) \leq 0$

Example 1: Monotone increasing (dose-response)

A dose-response relationship is naturally monotone increasing—higher doses should not decrease the response.

Simulate data

True function: $f(x) = 3(1 - e^{-5x})$, a saturating exponential.

```
df = CSV.read("data.csv", DataFrame)
x = df.x; y = df.y
n = nrow(df)
f_true = 3.0 .* (1.0 .- exp.(-5.0 .* x))
```

200-element Vector{Float64}:

```
0.0035813093739730517
0.020641315576114705
0.02346605588919992
0.033901484075778865
0.05864430830768941
0.10801952103761714
0.1159700766851749
0.4060464108647236
0.5120548610923702
0.5307120418037653
```

```
2.973608911275417
2.9750097790978067
2.975626021707947
2.975630618230438
2.9766280983687334
2.9769268870331884
2.9774612640729052
2.977972722359742
2.9786316912891824
```

Fit unconstrained GAM vs SCAM

```
m_gam = gam(@gam_formula(y ~ s(x, k=15, bs=:cr)), df)
m_scam = scam(@gam_formula(y ~ s(x, k=15, bs=:mpi)), df)
```

Generalized Additive Model

Formula: $y \sim 1$

Family: Normal
Link: IdentityLink
Method: GCV

Parametric coefficients:

	Coef.	Std. Error	t	Pr(> t)
(Intercept)	2.41623	0.0203741	118.59	<1e-99

Approximate significance of smooth terms:

Smooth	edf	Ref.df
s(x,bs=mpi)	3.42	14

R^2 (adj) = 0.877 Deviance explained = 87.9%
Scale est. = 0.0830 n = 200

Compare fitted values

```
yhat_gam = predict(m_gam)
yhat_scam = predict(m_scam)

rmse_gam = sqrt(mean((yhat_gam .- f_true).^2))
rmse_scam = sqrt(mean((yhat_scam .- f_true).^2))

println("RMSE (unconstrained GAM): ", round(rmse_gam, digits=4))
println("RMSE (SCAM, monotone increasing): ", round(rmse_scam, digits=4))
```

RMSE (unconstrained GAM): 0.0509
RMSE (SCAM, monotone increasing): 0.05

Verify monotonicity

The SCAM fit should be monotonically non-decreasing:

```

diffs = diff(yhat_scam)
println("Min successive difference (SCAM): ", round(minimum(diffs), digits=6))
println("All non-decreasing: ", all(diffs .>= -1e-10))

diffs_gam = diff(yhat_gam)
println("Min successive difference (GAM): ", round(minimum(diffs_gam), digits=6))
println("GAM all non-decreasing: ", all(diffs_gam .>= -1e-10))

```

```

Min successive difference (SCAM): 1.0e-6
All non-decreasing: true
Min successive difference (GAM): -0.001143
GAM all non-decreasing: false

```

Plot: GAM vs SCAM vs truth

```

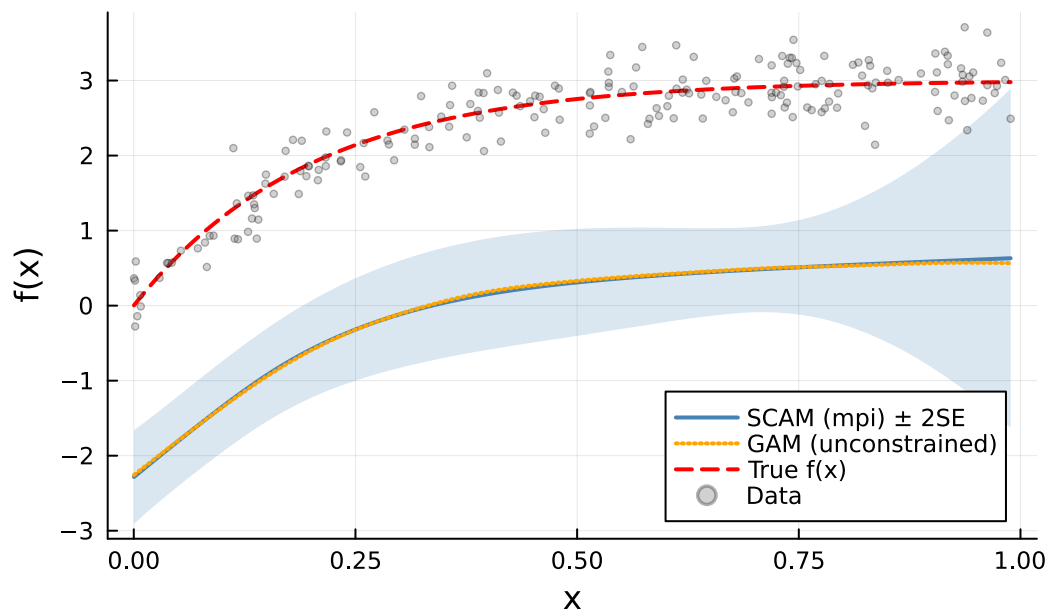
se_scam = smooth_estimates(m_scam; n=200)
x_grid = se_scam.covariates[:x]
f_hat_scam = se_scam.estimate
f_se_scam = se_scam.se
f_true_grid = 3.0 .* (1.0 .- exp.(-5.0 .* x_grid))

se_gam = smooth_estimates(m_gam; n=200)
f_hat_gam = se_gam.estimate

p = plot(x_grid, f_hat_scam;
    ribbon=2 .* f_se_scam, fillalpha=0.2, fillcolor=:steelblue,
    label="SCAM (mpi) ± 2SE", linewidth=2, color=:steelblue,
    xlabel="x", ylabel="f(x)",
    title="Monotone Increasing: SCAM vs GAM vs Truth",
    legend=:bottomright)
plot!(p, x_grid, f_hat_gam;
    label="GAM (unconstrained)", linewidth=2, color=:orange, linestyle=:dot)
plot!(p, x_grid, f_true_grid;
    label="True f(x)", linewidth=2, color=:red, linestyle=:dash)
scatter!(p, x, y;
    label="Data", alpha=0.3, markersize=2, color=:grey40)
p

```

Monotone Increasing: SCAM vs GAM vs Truth



Example 2: Convex function

Cost functions and accelerating growth curves are often convex.

Simulate data

True function: $f(x) = 2x^2$

```
df_cx = CSV.read("data_cx.csv", DataFrame)
x_cx = df_cx.x
y_cx = df_cx.y

df2 = DataFrame(y=y_cx, x=x_cx)
f_true2 = 2.0 .* x_cx.^2
```

200-element Vector{Float64}:

```
1.0272883703943119e-6
3.432112166702789e-5
4.439952989195642e-5
9.299474682423356e-5
0.0002806088263862691
```

```
0.0009682147886231632
0.0011190438882596383
0.01522772751099268
0.0252194799807833
0.027288906139353326
```

```
16.131253365630887
16.50515289191781
16.677748261268793
16.679055387058067
16.969970851777763
17.06003934671803
17.22468501346472
17.386733601386684
17.602323327505694
```

Fit with convexity constraint

```
m_cx = scam(@gam_formula(y ~ s(x, k=15, bs=:cx)), df2)

yhat_cx = predict(m_cx)
rmse_cx = sqrt(mean((yhat_cx .- f_true2).^2))
println("RMSE (convex SCAM): ", round(rmse_cx, digits=4))
```

RMSE (convex SCAM): 6.7682

Verify convexity

For a convex function, second differences should be non-negative:

```
first_diffs = diff(yhat_cx)
second_diffs = diff(first_diffs)
println("Min second difference: ", round(minimum(second_diffs), digits=6))
println("All convex: ", all(second_diffs .>= -1e-10))
```

Min second difference: -0.058956
All convex: false

Plot: Convex fit and second derivative

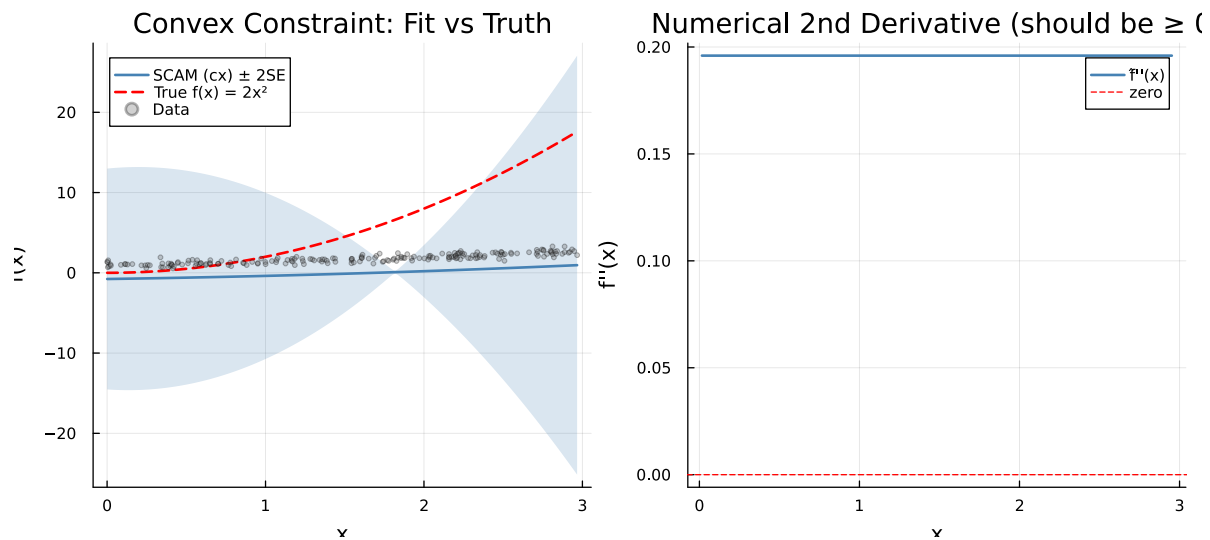
```
se_cx = smooth_estimates(m_cx; n=200)
x_grid_cx = se_cx.covariates[:x]
f_hat_cx = se_cx.estimate
f_se_cx = se_cx.se
f_true2_grid = 2.0 .* x_grid_cx.^2

p1 = plot(x_grid_cx, f_hat_cx;
    ribbon=2 .* f_se_cx, fillalpha=0.2, fillcolor=:steelblue,
    label="SCAM (cx) ± 2SE", linewidth=2, color=:steelblue,
    xlabel="x", ylabel="f(x)",
    title="Convex Constraint: Fit vs Truth",
    legend=:topleft)
plot!(p1, x_grid_cx, f_true2_grid;
    label="True f(x) = 2x²", linewidth=2, color=:red, linestyle=:dash)
scatter!(p1, x_cx, y_cx;
    label="Data", alpha=0.3, markersize=2, color=:grey40)

# Verify second derivative 0
dx = diff(x_grid_cx)
first_deriv = diff(f_hat_cx) ./ dx
x_mid = (x_grid_cx[1:end-1] .+ x_grid_cx[2:end]) ./ 2
dx2 = diff(x_mid)
second_deriv = diff(first_deriv) ./ dx2
x_mid2 = (x_mid[1:end-1] .+ x_mid[2:end]) ./ 2

p2 = plot(x_mid2, second_deriv;
    label="f''(x)", linewidth=2, color=:steelblue,
    xlabel="x", ylabel="f''(x)",
    title="Numerical 2nd Derivative (should be 0)",
    legend=:topright)
hline!(p2, [0.0]; label="zero", color=:red, linestyle=:dash, linewidth=1)

plot(p1, p2; layout=(1, 2), size=(900, 400))
```

Example 3: Monotone increasing and concave (diminishing returns)

Many real-world relationships show diminishing returns: the function increases but at a decreasing rate. This corresponds to a monotone increasing and concave constraint.

Simulate data

True function: $f(x) = 3\sqrt{x}$

```
df_micv = CSV.read("data_micv.csv", DataFrame)
x_micv = df_micv.x
y_micv = df_micv.y

df3 = DataFrame(y=y_micv, x=x_micv)
f_true3 = 3.0 .* sqrt.(x_micv)
```

```
200-element Vector{Float64}:
 0.28433012053092904
 0.3147719315378547
 0.4367684974148984
 0.6450893906079926
 0.6710266215708789
 0.7191942516143257
 0.7684079109443924
 0.7961350004577783
```

0.8628160497074876
0.863633819984572

2.9213968959339702
2.9269944201302573
2.931574047285353
2.932008562111724
2.936811966893178
2.9438191889181957
2.946733349169081
2.9558267325908676
2.988406288812731

Fit with monotone increasing + concave constraint

```
m_micv = scam(@gam_formula(y ~ s(x, k=15, bs=:micv)), df3)

yhat_micv = predict(m_micv)
rmse_micv = sqrt(mean((yhat_micv .- f_true3).^2))
println("RMSE (monotone increasing + concave): ", round(rmse_micv, digits=4))
```

RMSE (monotone increasing + concave): 1.5212

Verify constraints

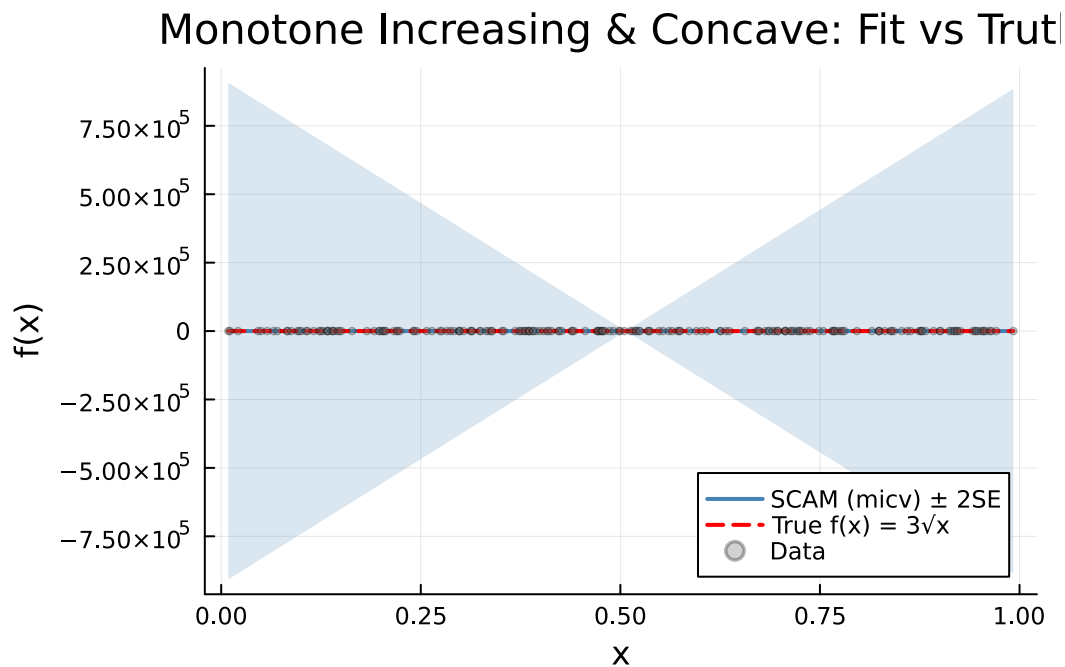
```
first_diffs_micv = diff(yhat_micv)
second_diffs_micv = diff(first_diffs_micv)
println("Min first difference (monotonicity): ", round(minimum(first_diffs_micv), digits=6))
println("Max second difference (concavity): ", round(maximum(second_diffs_micv), digits=6))
println("Monotone increasing: ", all(first_diffs_micv .>= -1e-10))
println("Concave: ", all(second_diffs_micv .<= 1e-10))
```

Min first difference (monotonicity): 8.0e-6
Max second difference (concavity): 0.01685
Monotone increasing: true
Concave: false

Plot: Monotone increasing & concave fit

```
se_micv = smooth_estimates(m_micv; n=200)
x_grid_micv = se_micv.covariates[:x]
f_hat_micv = se_micv.estimate
f_se_micv = se_micv.se
f_true3_grid = 3.0 .* sqrt.(x_grid_micv)

p = plot(x_grid_micv, f_hat_micv;
  ribbon=2 .* f_se_micv, fillalpha=0.2, fillcolor=:steelblue,
  label="SCAM (micv) ± 2SE", linewidth=2, color=:steelblue,
  xlabel="x", ylabel="f(x)",
  title="Monotone Increasing & Concave: Fit vs Truth",
  legend=:bottomright)
plot!(p, x_grid_micv, f_true3_grid;
  label="True f(x) = 3√x", linewidth=2, color=:red, linestyle=:dash)
scatter!(p, x_micv, y_micv;
  label="Data", alpha=0.3, markersize=2, color=:grey40)
p
```



SCAM fitting details

The `scam` function accepts the same arguments as `gam`, with additional control via `scam_control`:

```
ctrl = scam_control(  
  epsilon=1e-7,      # convergence tolerance  
  maxit=200,         # max Newton iterations  
  outer_maxit=200,   # max outer (smoothing parameter) iterations  
  trace=false,       # print iteration progress  
  gamma=1.0,         # GCV inflation factor  
  not_exp=false      # use exp() transform (default); true for softplus  
)  
  
m_ctrl = scam(@gam_formula(y ~ s(x, k=15, bs=:mpi)), df; control=ctrl)
```

Generalized Additive Model

Formula: $y \sim 1$

Family: Normal

Link: IdentityLink

Method: GCV

Parametric coefficients:

	Coef.	Std. Error	t	Pr(> t)
(Intercept)	2.41623	0.0203741	118.59	<1e-99

Approximate significance of smooth terms:

Smooth	edf	Ref.df
s(x,bs=mpi)	3.42	14

R^2 (adj) = 0.877 Deviance explained = 87.9%
Scale est. = 0.0830 n = 200

If no shape-constrained basis types are detected, `scam` automatically falls back to standard `gam` fitting:

```
m_fallback = scam(@gam_formula(y ~ s(x, k=15, bs=:cr)), df)
```

Generalized Additive Model

Formula: $y \sim 1$

Family: Normal

Link: IdentityLink

Method: GCV

Parametric coefficients:

	Coef.	Std. Error	t	Pr(> t)
(Intercept)	2.41623	0.0204657	118.06	<1e-99

Approximate significance of smooth terms:

Smooth	edf	Ref.df
s(x,bs=cr)	4.80	14

R^2 (adj) = 0.876 Deviance explained = 87.9%
Scale est. = 0.0838 n = 200

Comparing all constraint types

Let's fit each basis type to appropriate test data:

```
Random.seed!(42)
x_test = sort(rand(200))

# Monotone increasing: f(x) = x^2 (increasing on [0,1])
y_mpi = x_test.^2 .+ 0.1 .* randn(200)
# Monotone decreasing: f(x) = 1 - x^2
y_mpd = (1.0 .- x_test.^2) .+ 0.1 .* randn(200)
# Convex: f(x) = x^2
y_cx = x_test.^2 .+ 0.1 .* randn(200)
# Concave: f(x) = sqrt(x)
```

```

y_cv = sqrt.(x_test) .+ 0.1 .* randn(200)

basis_types = [:mpi, :mpd, :cx, :cv, :micx, :micv, :mdcx, :mdcv]
y_data = Dict(
    :mpi => y_mpi, :mpd => y_mpd, :cx => y_cx, :cv => y_cv,
    :micx => y_cx, :micv => y_cv, :mdcx => y_mpd, :mdcv => (1.0 .- sqrt.(x_test)) .+ 0.1 .* randn(200)
)

for bs in basis_types
    df_test = DataFrame(y=y_data[bs], x=x_test)
    m_test = scam(GamFormula(:y, Symbol[], true, [s(:x, k=10, bs=bs)]), df_test)
    yhat = predict(m_test)
    println("bs:=$bs - EDF: $(round(sum(m_test.edf), digits=2)), range: [$(round(minimum(yhat), digits=2)), $(round(maximum(yhat), digits=2))]")
end

```

```

bs=:mpi - EDF: 3.0, range: [-0.04, 1.0]
bs=:mpd - EDF: 3.04, range: [0.02, 1.06]
bs=:cx - EDF: 1.0, range: [-0.02, 0.99]
bs=:cv - EDF: 2.19, range: [0.13, 0.98]
bs=:micx - EDF: 1.0, range: [-0.02, 0.99]
bs=:micv - EDF: 2.42, range: [0.13, 0.98]
bs=:mdcx - EDF: 1.0, range: [0.14, 1.19]
bs=:mdcv - EDF: 1.0, range: [-0.05, 0.7]

```

Plot: All constraint types

```

true_funcs = Dict(
    :mpi => x_test.^2,
    :mpd => 1.0 .- x_test.^2,
    :cx => x_test.^2,
    :cv => sqrt.(x_test),
    :micx => x_test.^2,
    :micv => sqrt.(x_test),
    :mdcx => 1.0 .- x_test.^2,
    :mdcv => 1.0 .- sqrt.(x_test)
)

constraint_labels = Dict(
    :mpi => "Monotone ↑",
    :mpd => "Monotone ↓",

```

```

:cx  => "Convex",
:cv  => "Concave",
:micx => "Mono ↑ + Convex",
:micv => "Mono ↑ + Concave",
:mdcx => "Mono ↓ + Convex",
:mdcv => "Mono ↓ + Concave"
)

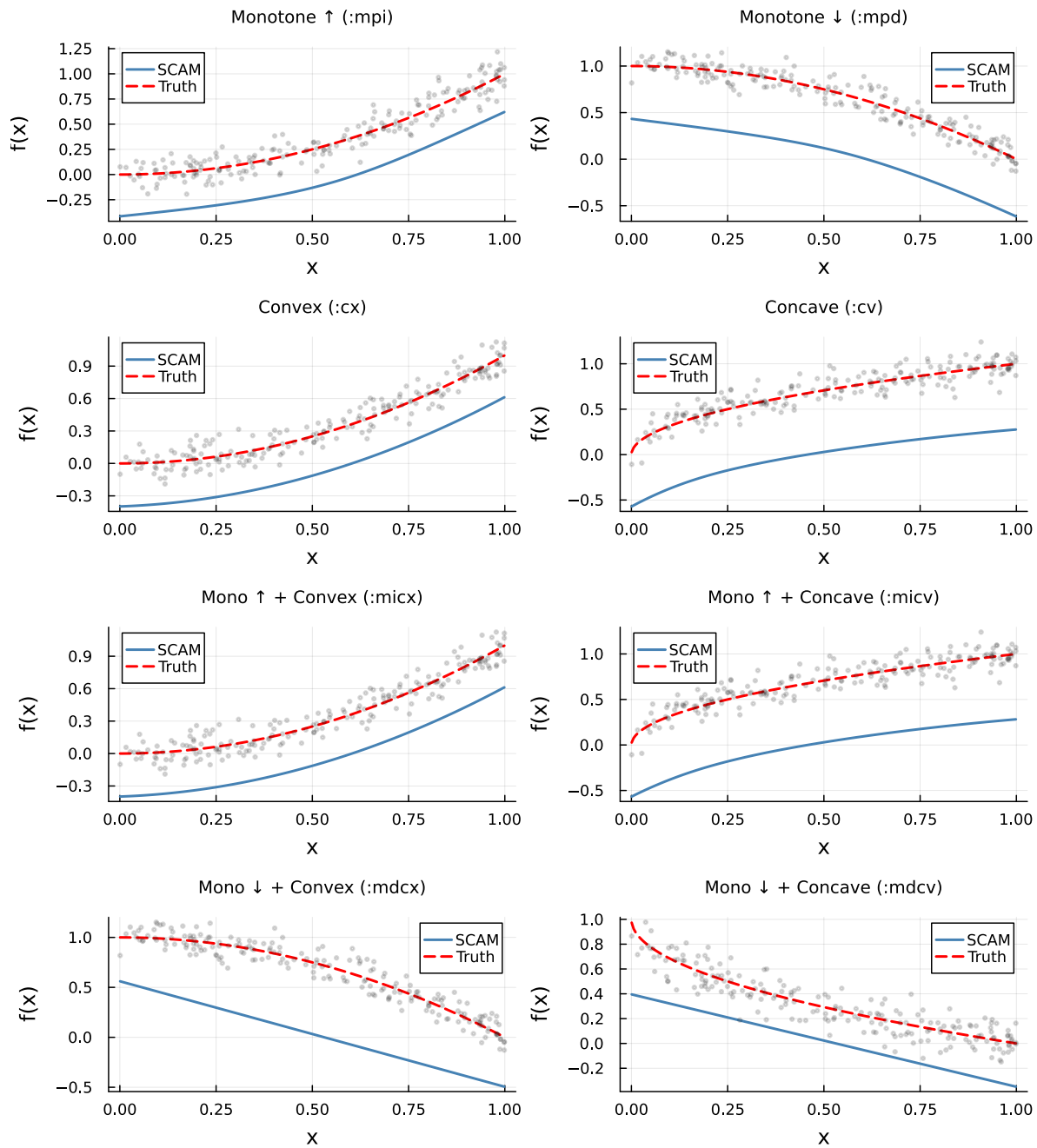
plots = []
for bs in basis_types
  df_test = DataFrame(y=y_data[bs], x=x_test)
  m_test = scam(GamFormula(:y, Symbol[], true, [s(:x, k=10, bs=bs)]), df_test)
  se_test = smooth_estimates(m_test; n=200)
  x_g = se_test.covariates[:x]
  f_g = se_test.estimate

  f_true_g = if bs in [:mpi, :cx, :micx]
    x_g.^2
  elseif bs == :mpd
    1.0 .- x_g.^2
  elseif bs in [:cv, :micv]
    sqrt.(x_g)
  elseif bs == :mdcx
    1.0 .- x_g.^2
  else
    1.0 .- sqrt.(x_g)
  end

  pi = plot(x_g, f_g;
    label="SCAM", linewidth=2, color=:steelblue,
    title=constraint_labels[bs] * " (:$bs)",
    xlabel="x", ylabel="f(x)", legend=:best, titlefontsize=9)
  plot!(pi, x_g, f_true_g;
    label="Truth", linewidth=2, color=:red, linestyle=:dash)
  scatter!(pi, x_test, y_data[bs];
    label="", alpha=0.15, markersize=1.5, color=:grey40)
  push!(plots, pi)
end

plot(plots...; layout=(4, 2), size=(800, 900))

```



Summary

Feature	GAM.jl (<code>scam</code>)	R <code>scam</code>
Fitting function	<code>scam(formula, data)</code>	<code>scam(formula, data=dat)</code>
Monotone increasing	<code>bs=:mpi</code>	<code>bs="mpi"</code>
Monotone decreasing	<code>bs=:mpd</code>	<code>bs="mpd"</code>
Convex	<code>bs=:cx</code>	<code>bs="cx"</code>
Concave	<code>bs=:cv</code>	<code>bs="cv"</code>
Mono. inc. + convex	<code>bs=:micx</code>	<code>bs="micx"</code>
Mono. inc. + concave	<code>bs=:micv</code>	<code>bs="micv"</code>
Mono. dec. + convex	<code>bs=:mdcx</code>	<code>bs="mdcx"</code>
Mono. dec. + concave	<code>bs=:mdcv</code>	<code>bs="mdcv"</code>
Control parameters	<code>scam_control()</code>	<code>scam.control()</code>

Shape constraints are enforced through the SCOP-spline reparameterization using the exponential function, ensuring constraints hold exactly at the spline knots and approximately between them.